

Project Design Phase-II

Technology Stack (Architecture & Stack)

Medical Diagnosis Center by Service Portal (ServiceNow)

Date	02 October 2026
Project Name	Medical Diagnosis Center by Service Portal (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture:

The architecture is centered on the ServiceNow cloud platform. The patient/requester accesses the diagnostic test service through Service Portal and Service Catalog. Catalog variables and configured form behaviour capture patient, test and appointment information. Flow Designer receives the submitted Requested Item context, retrieves the catalog variables, creates the configured diagnostic booking/operational record, validates or routes approval where configured, evaluates the approval result, updates the record and sends lifecycle notifications. Roles and ACLs provide access control, while Flow Designer execution details and platform debugging outputs support troubleshooting.

Architecture / Workflow Diagram

Revised for clear separation of diagnostic booking stages, decision branches and supporting ServiceNow capabilities.

1. Service Catalog	2. Get Variables	3. Create Booking	4. Approval*	5. Evaluate	6. Update & Notify
Diagnostic request	Patient/test/slot values	Booking record	Configured approval	Approved / rejected	Status + notification

* Approval is applied where the diagnostic request is configured to require approval.

Infrastructure Demarcation

- Cloud / Platform: ServiceNow cloud instance (Personal Developer Instance during development/training).
- Client side: Web browser accessing Service Portal; no separate native mobile application is required for the documented solution.
- Application logic: ServiceNow configuration and low-code automation, primarily Service Catalog, UI Policies, catalog variables and Flow Designer.
- Data storage: ServiceNow Request / Requested Item records plus the configured diagnostic booking/operational record used by the current implementation.
- External interfaces: No production laboratory-device API, payment gateway, SMS, WhatsApp or third-party integration is claimed as implemented in the current version.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Patient-facing booking and appointment-status experience.	ServiceNow Service Portal; Service Catalog; catalog variables
2	Form Behaviour	Guided diagnostic-test booking experience, visibility and mandatory-field behaviour.	ServiceNow UI Policies; Catalog Client Scripts where required
3	Patient & Test Data Capture	Structured patient, diagnostic-test and appointment information used by downstream automation.	Service Catalog variables; mandatory fields; configured catalog-item structure
4	Workflow Automation	End-to-end orchestration of diagnostic request processing, record creation, approval and branching.	ServiceNow Flow Designer; triggers; actions; conditions
5	Approval Management	Approval request, decision capture and approved/rejected branching for configured diagnostic requests.	Flow Designer Ask for Approval; approval rules; approval status condition
6	Diagnostic Operational Data	Structured booking/diagnostic record created and updated from submitted catalog values.	ServiceNow configured operational table / record
7	Platform Records	Native request context generated from Service Catalog submission.	ServiceNow Request / Requested Item (REQ / RITM) records
8	Notifications	Booking-created, approval/status and report-availability communications.	ServiceNow Flow Designer Send Email action / configured notifications
9	Security	Controlled access to patient, appointment and diagnostic information.	ServiceNow Roles; Access Control Lists (ACLs)
10	Diagnostics	Trace workflow execution, mappings, conditions and failures.	Flow Designer Execution Details; platform logs / debugging outputs
11	File / Report Storage	Support report availability and document handling within the documented ServiceNow scope.	ServiceNow-managed platform storage as applicable; report record/document capability where configured
12	External API / Integration	Future integration point for laboratory systems, diagnostic devices, payment or external notification services.	Not implemented in current version; integration can be added as future scope

S.No	Component	Description	Technology
13	Machine Learning Model	No ML inference or model dependency is required by the current booking workflow.	Not applicable to the current version
14	Infrastructure (Server / Cloud)	Application execution and hosting environment.	ServiceNow cloud instance / Personal Developer Instance for development and training

Technology Flow

Browser → Service Portal → Diagnostic Test Catalog Item → Patient/Test Variables & UI Policies → Flow Designer → REQ/RITM + Configured Diagnostic Booking Record → Approval (where configured) → Record Update → Notification → Diagnostic Fulfilment / Report Handoff → Patient Tracking.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology / Design Choice
1	Open-Source Frameworks	No separate open-source application framework is required by the documented core implementation.	ServiceNow platform components; no specific external framework asserted
2	Security	Protect patient, appointment and diagnostic records and administrative capabilities according to user role; enforce controlled read/write access.	ServiceNow roles, ACLs and platform authentication/session controls
3	Scalable Architecture	Configuration-driven design permits additional diagnostic categories, tests, variables, approval rules and fulfilment paths without replacing the complete solution.	ServiceNow modular Service Catalog + Flow Designer + configured record design
4	Availability	Application availability follows the availability of the deployed ServiceNow instance and related platform services.	ServiceNow-managed cloud platform; no separate availability layer
5	Performance	Minimize unnecessary client scripting and use Flow Designer for maintainable server-side orchestration; validate variable mapping and workflow execution for normal booking scenarios.	ServiceNow workflow execution + lightweight portal/client configuration
6	Maintainability	Use configuration-first components and document the setup so another administrator can reproduce or modify the solution.	Service Catalog, Flow Designer, UI Policies, ACLs and documented configuration
7	Observability / Troubleshooting	Trace workflow runs, action inputs/outputs, condition paths and record mappings when failures occur.	Flow Designer Execution Details; platform logs/debug outputs
8	Compatibility	Ensure the patient/requester experience operates through the intended Service Portal/catalog experience and does not depend on backend-only UI behaviour.	Service Portal + Service Catalog + supported ServiceNow configuration

Implemented vs Future Technology Boundary

- Implemented/documented core: Service Portal, Service Catalog, catalog variables, configured form behaviour, Flow Designer, diagnostic/operational record creation and update, approval handling, condition branching, email notifications, roles/ACLs and execution tracing.
- Future integrations: laboratory information systems, diagnostic-device interfaces, payment gateways, automated fulfilment integrations, external messaging channels and advanced diagnostic operational dashboards.
- Not applicable in the current version: a machine-learning inference model and a separately managed application server/database outside ServiceNow.